

Problem Set 1

Matthew Farah
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Problem 1:

Goofus, a MAT102 student, has asked you to critique his proof.

Proposition. Let $n \in \mathbb{Z}$. Then n is odd if and only if n^2 is odd.

Proof. n odd $\implies \exists k, n = 2k + 1 \implies n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2l + 1 \implies n^2$ odd.

n even $\implies n = 2k \implies n^2 = (2k)^2 = 4k^2 = 2(2k^2) = 2l \therefore n^2$ even.

Hence proved. □

Goofus' proof is logically correct, but very poorly communicated. Unfortunately, Goofus would get a score of 0/10 for submitting this proof in MAT301. Nice try, Goofus!

(a) Point out five things in Goofus' proof that need improvement.

Solution:

(b) Write your own proof of this proposition, focusing on improving the communication issues you noticed in Goofus' proof.

Solution:

Problem 2:

Consider the following binary operation on $\mathbb{Q}$:

$$x \otimes y := x + y + xy$$

For example, $2 \otimes 3 = 11$.

(a) Prove that $\otimes$ is associative.

Solution:

Let a, b , and $c \in \mathbb{Q}$ be arbitrary. To prove associativity, we must show that $(a \otimes b) \otimes c = a \otimes (b \otimes c)$. By simply using the given definition of $\otimes$ and basic facts related to multiplication and addition on $\mathbb{Q}$, we can indeed see that:

$$\begin{aligned}
 (a \otimes b) \otimes c &= (a + b + ab) \otimes c && \text{(By definition of } \otimes \text{)} \\
 &= a + b + ab + c + (a + b + ab)c && \text{(By definition of } \otimes \text{)} \\
 &= a + b + ab + c + ac + bc + abc && \text{(By distributivity of multiplication over addition on } \mathbb{Q} \text{)} \\
 &= a + ab + ac + abc + b + c + bc && \text{(By commutativity of addition on } \mathbb{Q} \text{)} \\
 &= a + ab + ac + abc + (b \otimes c) && \text{(By definition of } \otimes \text{)} \\
 &= a + a(b + c + bc) + (b \otimes c) && \text{(By distributivity of multiplication over addition on } \mathbb{Q} \text{)} \\
 &= a + a(b \otimes c) + (b \otimes c) && \text{(By definition of } \otimes \text{)} \\
 &= a + (b \otimes c) + a(b \otimes c) && \text{(By commutativity of addition on } \mathbb{Q} \text{)} \\
 &= a \otimes (b \otimes c) && \text{(By definition of } \otimes \text{)}
 \end{aligned}$$

And so, since $(a \otimes b) \otimes c = a \otimes (b \otimes c)$, $\otimes$ is associative over $\mathbb{Q}$.

(b) Prove that $\otimes$ is commutative.

Solution:

Let $a, b \in \mathbb{Q}$ be arbitrary. To prove commutativity, we must show that $a \otimes b = b \otimes a$, by a similar process as before, we can see that:

$$\begin{aligned}
 a \otimes b &= a + b + ab && \text{(By definition of } \otimes \text{)} \\
 &= b + a + ab && \text{(By commutativity of addition on } \mathbb{Q} \text{)} \\
 &= b + a + ba && \text{(By commutativity of multiplication on } \mathbb{Q} \text{)} \\
 &= b \otimes a && \text{(By definition of } \otimes \text{)}
 \end{aligned}$$

Thus, since $a \otimes b = b \otimes a$, $\otimes$ is indeed commutative over $\mathbb{Q}$.

(c) Find, with a proof, the identity element for $\otimes$.

Solution:

The identity element for $\otimes$ in $\mathbb{Q}$ is 0. To prove that 0 is really $\otimes$'s identity element, we need to show that for arbitrary $a \in \mathbb{Q}$, $a \otimes 0 = 0 \otimes a = a$. Letting a be any such element in $\mathbb{Q}$ then, expanding gives:

$$\begin{aligned}
\boxed{a \otimes 0} &= a + 0 + a(0) && \text{(By definition of } \otimes \text{)} \\
&= a + 0 + 0 && \text{(By property of multiplication by zero)} \\
&= \boxed{a} && \text{(Since 0 is the additive identity on } \mathbb{Q} \text{)} \\
&= 0 + a + 0 && \text{(Since 0 is the additive identity on } \mathbb{Q} \text{)} \\
&= 0 + a + 0(a) && \text{(By property of multiplication by zero)} \\
&= \boxed{0 \otimes a} && \text{(By definition of } \otimes \text{)}
\end{aligned}$$

Therefore, since 0 satisfies $a \otimes 0 = 0 \otimes a = a$ for arbitrary $a \in \mathbb{Q}$, 0 is the identity element of $\otimes$ over $\mathbb{Q}$.

(d) Find, with proof, the inverse of $\frac{5}{6}$.

Solution:

The inverse of $\frac{5}{6}$ is $\frac{-5}{11}$. To prove that this is indeed the case, we need to show that for $\frac{5}{6} \otimes \frac{-5}{11} = \frac{-5}{11} \otimes \frac{5}{6} = 0$, since we previously determined 0 to be the identity element of $\otimes$ over $\mathbb{Q}$. Expanding out, we find that:

$$\begin{aligned}
\boxed{\frac{5}{6} \otimes \frac{-5}{11}} &= \frac{5}{6} - \frac{5}{11} + \left(\frac{5}{6}\right) \left(\frac{-5}{11}\right) && \text{(By definition of } \otimes \text{)} \\
&= \frac{55}{66} - \frac{30}{66} - \frac{25}{66} && \text{(Simplifying)} \\
&= \boxed{0} && \text{(Simplifying)} \\
&= -\frac{30}{66} + \frac{55}{66} - \frac{25}{66} && \text{(Expanding)} \\
&= -\frac{-5}{11} + \frac{5}{6} + \left(\frac{-5}{11}\right) \left(\frac{5}{6}\right) && \text{(Expanding)} \\
&= \boxed{\frac{5}{6} \otimes \frac{-5}{11}} && \text{(By definition of } \otimes \text{)}
\end{aligned}$$

And so, $-\frac{5}{11}$ is the inverse of $\frac{5}{6}$ under $\otimes$ in $\mathbb{Q}$.

(e) Describe all rational numbers that are invertible with respect to $\otimes$. In fact, given a rational number $x \in \mathbb{Q}$, find a formula for the inverse of x with respect to $\otimes$.

Solution:

All rational numbers, except -1 , are invertible.

To see this clearly, I will proceed by a direct proof. Letting $a \in \mathbb{Q}$ be arbitrary, we know definitionally that the inverse of a is an element $a^{-1} \in \mathbb{Q}$ satisfying $a \otimes a^{-1} = a^{-1} \otimes a = 0$, since 0 was determined earlier to be the identity element of $\otimes$ over $\mathbb{Q}$. By definition of $\otimes$, we therefore can see that $a \otimes a^{-1} = 0$ yields:

$$\begin{aligned} a \otimes a^{-1} &= 0 && \text{(By definition of inverse)} \\ \implies a + a^{-1} + aa^{-1} &= 0 && \text{(By definition of } \otimes \text{)} \\ \implies a + (1 + a)a^{-1} &= 0 && \text{(By distributivity of multiplication over addition on } \mathbb{Q} \text{)} \\ \implies (1 + a)a^{-1} &= -a && \text{(Moving } a \text{ to the right-hand side)} \\ \implies a^{-1} &= -\frac{a}{1+a} && \text{(Moving } a \text{ to the right-hand side)} \end{aligned}$$

Notice however that the final equality we arrived at is undefined in the case that $a = -1$, meaning that $-1 \in \mathbb{Q}$ cannot possibly have an inverse under $\otimes$. The equality however is perfectly valid for all other numbers in $\mathbb{Q}$. Thus, every element in $\mathbb{Q} \setminus \{-1\}$ has an inverse under $\otimes$, and for any $a \in (\mathbb{Q} \setminus \{-1\})$, its inverse is given by $a^{-1} = -\frac{a}{1+a}$.

Problem 3:

Let $2 \leq n$ and let $u \in \mathbb{Z}_n$. Recall that u is a “unit modulo n ” if it is multiplicatively invertible. Thus, there exists $v \in \mathbb{Z}_n$ such that $uv \equiv 1 \pmod{n}$. Let $\mathbb{Z}_n^\times$ denote the set of all unit modules of n ; For example, $\mathbb{Z}_9^\times = \{1, 2, 4, 5, 7, 8\}$.

(a) Write down the multiplication table for the group $\mathbb{Z}_{36}^\times$.

Solution:

First, we must actually determine $\mathbb{Z}_{36}^\times$. We know by part b.) of this problem (the instructor is allowing us to use part b.) to solve part a.)) that:

$$\gcd(0, 36) = 36 \quad \gcd(6, 36) = 6 \quad \gcd(12, 36) = 12 \quad \gcd(18, 36) = 18 \quad \gcd(24, 36) = 6 \quad \gcd(30, 36) = 6$$

$$\boxed{\gcd(1, 36) = 1} \quad \boxed{\gcd(7, 36) = 7} \quad \boxed{\gcd(13, 36) = 1} \quad \boxed{\gcd(19, 36) = 1} \quad \boxed{\gcd(25, 36) = 1} \quad \boxed{\gcd(31, 36) = 1}$$

$$\gcd(2, 36) = 2 \quad \gcd(8, 36) = 4 \quad \gcd(14, 36) = 2 \quad \gcd(20, 36) = 4 \quad \gcd(26, 36) = 2 \quad \gcd(32, 36) = 4$$

$$\gcd(3, 36) = 3 \quad \gcd(9, 36) = 3 \quad \gcd(15, 36) = 3 \quad \gcd(21, 36) = 3 \quad \gcd(27, 36) = 3 \quad \gcd(33, 36) = 3$$

$$\gcd(4, 36) = 4 \quad \gcd(10, 36) = 2 \quad \gcd(16, 36) = 4 \quad \gcd(22, 36) = 2 \quad \gcd(28, 36) = 4 \quad \gcd(34, 36) = 2$$

$$\boxed{\gcd(5, 36) = 1} \quad \boxed{\gcd(11, 36) = 1} \quad \boxed{\gcd(17, 36) = 1} \quad \boxed{\gcd(23, 36) = 1} \quad \boxed{\gcd(29, 36) = 1} \quad \boxed{\gcd(35, 36) = 1}$$